Response to the recommender and reviewers

Manuscript: *Cross-Cultural Validation and Ideological Fairness of a Historical Perspective Taking Instrument: Evidence from Czech Secondary Students*

Dear Recommender and Reviewers,

Thank you for the careful and constructive reviews. I revised the manuscript to sharpen its measurement focus, add the requested reporting, and bound its claims to the single scenario, item format, sample, and ideological range studied. While revising, I also rechecked the analyses and corrected several reported results; these corrections are listed under *Additional corrections* at the end of this letter.

Each comment is quoted below, followed by my response. Page and line numbers in the quoted comments refer to the submitted version. In my responses, section, table, and figure references refer to the revised manuscript and supplement, and page and line numbers refer to the clean revised PDF. A tracked-changes PDF against the submitted version accompanies this letter.

Location map for the revised files

Location	Contents
Abstract	Principal results and scope boundary
Sections 1–2	Background, prior evidence, and research questions RQ1 and RQ2
Sections 3–4	Method, psychometric evidence, and focal relationships
Section 5	Interpretation, limitations, and recommendations
Tables 1–5, Figure 1	Factor-model fit, descriptives, DIF, invariance, correlations; observed relationships
Tables S1–S3, S3b	Item parameters, loadings, reliability, pairwise sample sizes
Tables S4, S4b	Instrument source populations; analysis-specific sample sizes
Table S5	Preregistration mapping and deviations
Tables S6–S7	NS-only invariance fit; HPT composite by ideology tertile

Tables 2 and 3 of the submitted version (a loading summary and a reliability summary) were removed because they repeated the text and Tables S2–S3; the remaining tables were renumbered.

Recommender

Comment 1

> The biggest limitation to me is content validity of the instrument. The measure presents only one scenario from one historical and sociopolitical context. Although multiple items are rated about that scenario, it is difficult to know how much the content of

the scenario itself represents the underlying construct of interest. A measure in which these items were applied to multiple scenarios that span time and place would be much stronger. You list this as a limitation, but the limitation is currently included with sampling in one brief sentence. It should elaborated in terms of the more general issue of content validity.

Response: I agree that this is the main limitation and have made it the first limitation in Section 5.6 (pp. 36–37, lines 702–710), under the heading *Content validity*, separate from sampling. It states that nine ratings of one Weimar voting dilemma are repeated judgments about the same content rather than nine independent samples of the domain, so the scores risk construct underrepresentation (Messick, 1995). It explains that scenario-specific knowledge, familiarity, and moral salience cannot be separated from perspective taking, that the evidence supports interpretations of performance on this task rather than HPT as a general competence, and that the fairness findings are bounded in the same way. It calls for parallel item sets applied to several scenarios that vary in period, place, political valence, and familiarity. The Conclusion (Section 5.8, p. 39, lines 754–756) now opens with "For this single Weimar scenario".

Comment 2

You say discriminant validity is supported based on better fit of the three factor model and factor correlations below .85, but this is a weak standard. Evidence that the scales differentially predict external outcomes would be more convincing, and that should be listed as a question for further research.

Response: I replaced the former below-.85 rule with bounded internal-structure language. Section 4.1 (pp. 22–23, lines 458–463) reports the factor correlations as evidence that the item sets are distinguishable in this sample and states that this does not establish discriminant validity. Section 5.7 now lists, as recommendation (f) (p. 39, lines 750–752), testing whether the subscales differentially predict external criteria; protocol step 5 in Section 5.5 (p. 36, lines 698–699) makes the same point.

Comment 3

I recommend using construct names rather than scale abbreviations in table 7 and the results section to improve readability.

Response: Table 5 (formerly Table 7) and the Results narrative now use construct names; the table's rows are labelled "HPT composite", "Contextualization", "Present-oriented perspective (rev.)", and so on. The primary six-item composite is defined once in Section 3.3.1 (p. 16, lines 326–330) and called "HPT composite" thereafter. Scale abbreviations are retained only where they identify named instruments.

Comment 4

You should add confidence intervals to the effects presented in Table 7.

Response: Table 5 now reports Fisher-transformed 95% confidence intervals for every correlation. It states that the intervals are descriptive and do not adjust for clustering (p. 30, lines 582–587). Exact pairwise sample sizes are reported in Table S3b, and the values reproduce from `revision_reporting_analyses.R`.

Comment 5

> I agree with your framing of internal consistency for brief scales in the discussion. You could cite Loevinger's attenuation paradox rather than framing this as a deficiency relative to standard benchmarks. At the same time, you should acknowledge that this will constrain validity estimates.

Response: I adopted your suggestion. Section 5.3 (pp. 33–34, lines 638–654) now opens with the breadth-precision trade-off and Loevinger's attenuation paradox: raising reliability does not necessarily raise validity, and three items that sample different considerations about one decision can serve the construct better than near-duplicates. One sentence acknowledges that the resulting precision is modest and limits individual-level interpretation. The section then states the consequence you asked for: modest reliability attenuates validity coefficients, so the knowledge correlation is a lower bound, and the ideology associations are attenuated in the same way, which limits the strength of the null fairness evidence. Table S3 reports raw alpha with standard-error intervals, polychoric alpha, and omega for all scored constructs; the one-factor omega estimates for multidimensional composites are identified as descriptive.

Reviewer 1

Comment 1

> Does the abstract present the main findings of the study? [ ] Yes, [x] No

Response: The abstract (pp. 1–2, lines 10–22) now gives the primary knowledge and ideology correlations with 95% confidence intervals and states that no DIF was detected with this sample and procedure. It reports that multi-group CFA fit-index changes were compatible with scalar invariance and qualifies this evidence because the extreme groups are small and clustering was not modelled. Sections 4.9, 5.1, and 5.8 (pp. 29–30, lines 572–578; pp. 32–33, lines 622–625; p. 39, lines 760–762) report the authoritarianism–reversed-presentism association as an unadjusted, hypothesis-generating result that was not subjected to the preregistered multiplicity procedure.

Comment 2

> The instrument has already been used in the Dutch context, and also there issues emerged regarding the validity and reliability of the scale, and, for example, there was no significant correlation with school achievement in history.

Response: In this sample, history grade correlated with the HPT composite (r = -.15, p = .010; the negative sign reflects the Czech scale, on which 1 is the best grade). Section 4.9 (p. 29, lines 567–570) now notes that this resembles the original German result rather than the Dutch applications, in which HPT scores were not significantly correlated with history grades. The Dutch validity and reliability findings are now described in Section 2.1 (see Comment 7).

Comment 3

> I do, however, welcome the fact that the article provides a protocol for this type of cross-cultural validation. Maybe the focus could be more on that aspect of the paper than on the different pathways disucssed at page 34.

Response: I agree that the protocol is the more useful contribution. The adaptation evidence and staged fairness evaluation now organize the paper from the research questions through the Conclusion. The former discussion of pathways (three paragraphs in Section 5.5 and one in Section 5.4) is condensed into a single paragraph in Section 5.4 (p. 35, lines 669–679). It lists the explanations the data cannot distinguish (a genuine null, cancelling pathways, a format effect, and task-demand compliance) and points to the response-process evidence in limitation 8 that could separate them. I also removed the phrase "the instrument's resistance to attitudinal contamination", which overstated the evidence.

Comment 4

> Construct-irrelevant variance: the idea that test scores are influenced by respondents' positionality and could lead to two pathways: lower cognitive effort, making the instrument less effective in capturing a cognitive skill. Another interesting possibility is that the scores are influenced by authoritarianism. This latter aspect could be explained more clearly in the theoretical framework. What exactly makes the HPT instrument more difficult for these students? Perhaps the authors could also make the connection here with researchers who regard the moral and affective dimensions of HPT as an indispensable part of the construct, and the link with political attitudes could also be elaborated somewhat more explicitly on a conceptual level (also because of the instruments used in the methods

Response: Sections 2.1 and 2.3 now distinguish contextual reconstruction from attitude-congruent endorsement, reduced integrative complexity, task-demand shortcuts, moral evaluation, and affective engagement. A construct-boundary paragraph (p. 11, lines 220–227) explains when affect and moral engagement may be target-relevant under broad historical-empathy accounts and when option matching or attitude congruence is construct-irrelevant under the narrower cognitive score interpretation used here. Section 2.3 (p. 10, lines 198–204) now also explains concretely why the ideology question arises for this instrument: every contextualization option describes Hannes as likely to vote for the NSDAP and every present-oriented option describes him as rejecting it, so the POP–CONT continuum coincides with the direction of the political choice (Wilschut & Schiphorst, 2019).

Comment 5

> On page 2, refer to the Hartmann & Hasselhorn publication instead of, or in combination with, the German-language dissertation (in APA format).

Response: The manuscript now cites Hartmann and Hasselhorn (2008) throughout. The Czech version was translated from that published English article, as the submitted version stated; Section 3.1 (p. 13, lines 268–272) says so and mentions Hartmann's (2008) dissertation once, as the instrument's original development source. Section 3.3.1 (pp. 15–16, lines 316–319) and the adaptation record in the reproducibility package were made consistent with this description.

Comment 6

> Pp. 5 and 6: use Hartmann & Hasselhorn (2008) and Lee & Ashby (2001) instead of the titles.

Response: Corrected. These citations now render as author-date citations, Hartmann and Hasselhorn (2008) and Lee and Ashby (2001), rather than as publication titles.

Comment 7

> L. 41: "The instrument developed by *Perspektivenübernahme als eine Kompetenz historischen Verstehens* (2008) is among the most extensively validated standardized HPT measures in European research." I am not sure this is true. I know of some Turkish work and some work in the Netherlands. The literature contains only a small, methodologically heterogeneous follow-up tradition surrounding the HPT of Hartmann and Hasselhorn. The German standardization positions historical perspective taking as a context-dependent competence, whereas Turkish and Dutch-oriented applications mainly reveal the transferability-but also the vulnerability-of the measurement. Some studies report validity issues and indicate that the method of measurement cannot always be transferred to other substantive contexts.

Response: Section 2.1 (pp. 6–7, lines 124–140) now distinguishes the original German PCA result from the separate latent-class analysis and from the correlated three-factor CFA tested here. It reports successful Dutch reuse of the Nazi scenario while noting weaker reliability and validity evidence for transfer to a slavery scenario and later evidence that students sometimes misunderstood response options. Section 2.1 also cites Wilschut and Schiphorst (2019) as a second transfer study: with two new scenarios and 69 Dutch Grade 10 students, principal components did not recover the three response categories and total-score reliability was low. Claims that the instrument had been extensively validated or that the Czech CFA replicated an original three-factor structure were removed. The Turkish work is addressed under Comment 16.

Comment 8

> Page 3, l. 61: "The scenario describes a young man in the Weimar Republic deliberating whether to join an extremist political movement." I believe this is not correct. The original scenario by Hartmann & Hasselhorn states: "The scenario we developed presents a young man (Hannes) in the Weimar Republic in Germany, thinking about which party to vote for in the 1930 election" (p. 266). Voting is something different from joining. Has the scenario been changed for the Czech audience? Is this an example of the "cross-cultural adaptation" of this item?

Response: You are right, and the scenario was not changed for the Czech version. I compared the German source, published English article, and administered Czech instrument. Sections 1 and 3.3.1 (pp. 3–4, lines 61–64; pp. 15–16, lines 316–319) now state that Hannes discusses Germany's crisis and the 1930 election and that students judge explanations including whether he could vote for an anti-democratic party such as the NSDAP. The manuscript no longer describes joining an extremist movement, and the illustrative statement in Section 2.3, which was not an instrument item, was replaced by a description of the actual option alignment (p. 10, lines 198–204; see Comment 4).

Comment 9

> Could it be possible to provide more information about the history education context in the Czech Republic? Is HPT part of the curriculum? Is history a compulsory subject in school, etc.?

Response: Section 2.2 (pp. 8–9, lines 169–176) now describes the lower-secondary framework's coverage of the 1920s and 1930s and its source-use expectation, with page-specific citation to the 2023 framework, and states that HPT is not named as a separate competence and that curriculum documents establish intended rather than enacted provision. Because no equivalent upper-secondary curriculum-document audit was conducted, the documentary claim is limited to the lower-secondary subsample and distinguished from teachers' eligibility confirmation at both levels.

Comment 10

> Methods: p. 15, could it be possible to report the internal consistency of these scales as well? I assume these are validated scales. Were they suitable for this target group?

Response: Section 4.3 (p. 23, lines 472–477) and Table S3 report reliability for FR-LF, KSA-3, historical knowledge, SDR-5, and the combined ideology score. Table S4 audits the original source populations. The manuscript now states (p. 17, lines 347–352) that the German FR-LF surveys included respondents aged 14 and above but were predominantly adult, while KSA-3 and SDR-5 were developed with adults; none directly validates a Czech adolescent version.

Comment 11

> Results: As a non-statistical expert, I think the authors did a good job of presenting the results. Perhaps there could be more focus on the main outcomes, and some tables could be moved to an appendix.

Response: A short roadmap at the opening of Results (p. 21, lines 425–429) now maps Sections 4.1–4.5 to RQ1, Sections 4.6–4.8 to RQ2, and Sections 4.9–4.10 to the focal relationships. Subsection openings state the principal result before technical detail. Detailed item loadings, reliability estimates, source populations, and analysis-specific sample sizes are in the supplement; Table 5 and Figure 1 foreground the principal relationships.

Comment 12

> P. 36, l. 680: I believe Huijgen already used such an approach; comparing a closed and an open format (and thinking aloud).

Response: Section 2.1 (p. 7, lines 137–139) now separates Huijgen and colleagues' transfer of the item-rating approach across topics from their later combination of closed ratings with think-alouds. I do not claim that the cited studies directly compared open and closed response formats. Limitation 8 in Section 5.6 (p. 38, lines 734–743) proposes that direct comparison, adding that written explanations have detected change that closed ratings in this format missed (Wilschut & Schiphorst, 2019).

Comment 13

> P. 37, l. 708: "gender…samples", please provide a reference for this claim.

Response: I found no direct source supporting the earlier adolescent gender-moderation statement and removed it. Section 5.6 (p. 37, lines 720–722) now states only that the sample imbalance limits assessment of gender heterogeneity and that no gender interaction or gender DIF analysis was conducted.

Comment 14

It is interesting that the issues with reliability are still present.

Response: As you note, reliability problems recur. Huijgen et al. (2014) and Wilschut and Schiphorst (2019) report weaker reliability when the format is moved to new content, and Section 5.3 (p. 34, lines 645–649) now places the Czech estimates against these: the Czech composite estimates (omega = .66–.70) are modest but higher than the total-score alphas of .22–.50 in the new-content adaptations, consistent with a structure that depends on the original scenario.

Comment 15

P. 34, regarding the different pathways: wouldn't it be an option to investigate these pathways in a more qualitative manner (thinking aloud?).

Response: Following your suggestion, limitation 8 in Section 5.6 (p. 38, lines 734–743) now specifies observable think-aloud indicators: use of contextual evidence and integration of conflicting considerations, moral evaluation and affective engagement, paraphrase or option matching, and explicit attitude-congruent endorsement. These indicators would help distinguish intended reconstruction from the competing pathways that Section 5.4 lists.

Comment 16

L. 45: "employed in comparative studies in historical reasoning" (please name those studies).

Response: Section 2.1 (p. 7, lines 131–145) names and describes the verified Huijgen studies and Wilschut and Schiphorst (2019) rather than referring generically to comparative studies. Claims unsupported by direct use of the instrument were removed. I also cite the Turkish qualitative case study you suggested (Aktın et al., 2026) after full-text verification: it involved sixth- and eighth-grade students and used the Hartmann and Hasselhorn components as deductive analytic categories. I do not treat it as evidence of direct nine-item administration or psychometric performance.

Reviewer 2

Comment 1

Are the research questions/hypotheses/predictions clearly presented? [ ] Yes, [X] No

Response: Section 2.5 (pp. 12–13, lines 253–264) now presents two explicit research questions, labelled RQ1 (psychometric evidence for the Czech adaptation) and RQ2 (whether scores are affected by political attitudes). The labels are reused in the Results roadmap and the Summary of Findings (Section 5.1). A concise disclosure identifies post-registration additions and omitted registered tests; Table S5 provides the full mapping of the six registered hypotheses.

Comment 2

First, in describing the participants on pg. 13, I would provide more descriptive data on the students and schools. For instance, what is the ideological distribution of the students? It is only mentioned in the Discussion section (which 17% of students having above average right-wing sympathies), but should be included earlier. It's the key independent variable being examined, so it needs more discussion of its distribution up

front. In addition, there is a mention of the most recent history grade, but more detail on the distribution of students' historical knowledge would be good to have. In addition, what is the socioeconomic background of the schools sampled? Given that this is not a representative sample, that would be a good statistic to include, in addition to the region and funding source of the schools.

Response: Section 3.2 (p. 15, lines 309–313) now gives participant-level ideology and knowledge medians alongside school distribution, gender, and grade level; Section 4.4 (p. 24, lines 484–490) provides the fuller distributions and floor and ceiling percentages. No defensible socioeconomic measure was collected. Sections 3.2 and 5.6 (p. 37, lines 723–726) state that socioeconomic coverage and associated selection bias cannot be characterized rather than inferring status from school location. Table S4b reports analysis-specific sample sizes.

Comment 3

Second, and this could be due to my lack of technical knowledge, but adding some simpler presentations of the relationship between ideology and HPT, and historical knowledge and HPT, would help (in addition to, not in replacement of, the great work done in the Methods section). There are CFA scores, DIF scores, betas from regression models, etc. But just add something more descriptive as well - a mean score for each ideology group, a data plot with a fit line illustrating the relationshiop between HPT and ideology and/or knowledge, etc. Adding a more 'naïve' or baseline representation of these key relationships could help orient the reader as they dig into the more technical statistical analysis.

Response: Table 5 reports unadjusted correlations with confidence intervals, and Figure 1 displays the continuous ideology-HPT and knowledge-HPT relationships with sample sizes and uncertainty. I also added Table S7, which reports the mean HPT composite with a 95% confidence interval for the low, middle, and high tertiles of the composite ideology score; a sentence in Section 4.9 (p. 30, lines 588–590) points to it. The continuous display remains primary because tertiles are arbitrary cut-points of a continuous score. The Results distinguish these descriptive patterns from the adjusted multilevel models.

Additional corrections

Changes to reported results since the submitted version

While revising, I rechecked the analyses against the archived data and code. The table lists every change to a reported result or disclosure relative to the submitted version.

Item	Submitted version	Revised version
Three-factor CFA fit	CFI .999, RMSEA .011 (unscaled)	CFI .975, RMSEA .039 (scaled WLSMV); described as good, not near-perfect
Bifactor vs. three-factor	ΔCFI .001, "no meaningful improvement"	ΔCFI +.018; fit favors the bifactor model; structural claims softened
MG-CFA admissibility	Heywood cases in the Low group	All solutions admissible; Heywood claim withdrawn

Item	Submitted version	Revised version
Focal FR-LF coefficient	$\beta = .012$, $p = .863$	$\beta = .016$, 95% CI [-.117, .149], $p = .817$, $n = 282$ (one scoring rule applied throughout); singular fit disclosed
Design-effect argument	DEFF 1.56/2.16 used to justify inference	Removed (it combined classroom size with school ICCs)
ICCs	School level only	School / class-within-school / total
DIF description	Uniform and non-uniform tests	Omnibus joint test; raw POP in the GRM disclosed
Modification index	Read as loading non-invariance	Read as a POP2 cross-loading diagnostic only
Authoritarianism-presentism	Not reported	Reported as exploratory and unadjusted
Preregistration disclosure	TOP form: one deviation (TOST bound)	Table S5 lists omitted registered analyses, post-registration additions, and the extension of the registered tertile-group MG-CFA from the CONT items to all nine items
Repository	OSF project	Zenodo, https://doi.org/10.5281/zenodo.23494026

None of these changes alters the substantive conclusion: historical knowledge remains the only consistent predictor, and no ideology effect or DIF was detected. Two corrections deserve particular mention. First, the submitted TOP form listed a single deviation from the preregistration; that statement was incomplete. Table S5 now discloses every omitted registered analysis (including the MIMIC model, the knowledge moderation tests, the full covariate set, one-tailed tests, and the Benjamini-Hochberg correction) and every post-registration addition, and the revised TOP form refers to it. Second, because OSF has discontinued project storage, the data, code, and documentation now reside in a Zenodo record; the preregistration remains at https://osf.io/zsngy.

Sincerely,

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